

Malhar Gudekar

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EDUCATION

University of Illinois Urbana Champaign

August 2024 - May 2026

Master of Science, Information Management

GPA: 3.8/4

• Coursework: Web Programming, Machine Learning & Cloud, Information Management, Information Consulting, Methods of Data Science

University of Mumbai

June 2019 - May 2023

Bachelor of Engineering, Computer Science

GPA: 9.25/10

• Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Object-Oriented Programming, Machine Learning, NLP

SKILLS

• **Languages & DSA:** Python, SQL, Data Structures & Algorithms (Arrays, Hashing, Trees, Graphs, Dynamic Programming), LeetCode

• **Backend & Systems:** FastAPI, REST APIs, Microservices, Distributed Systems, ETL Pipelines, Kafka, PySpark

• **Databases, Cloud & Tools:** PostgreSQL, Neo4j, AWS (S3, EC2, EKS), GCP, Docker, Git

WORK EXPERIENCE

Business Intelligence Group

August 2025 - December 2025

Technical Consultant

Champaign, USA

- Built and deployed production-grade LLM applications in Python, including multi-stage prompt pipelines, tool-integrated agentic workflows, and RAG patterns with vector embeddings, served via FastAPI with PostgreSQL for scalable enterprise cloud deployment
- Designed and iterated on LLM retrieval components, chunking strategies, embedding generation, context assembly, and reranking, optimizing prompt pipeline performance for low-latency RAG in production enterprise environments
- Improved LLM application performance by 64% through iterative component tuning and retrieval optimization, validated through structured testing practices and code review cycles to ensure reliability and accuracy of AI outputs
- Developed backend services enabling real-time, high-performance **AI model serving** of unstructured data across distributed systems

University of Illinois Urbana Champaign

May 2025 - Present

Research Assistant - iSchool

Champaign, USA

- Owned **end-to-end design, development, and integration** of cross-platform **AI backend system** ingesting wearable data for 100+ users, enabling scalable data collection and downstream analytics
- Designed and optimized **PostgreSQL schema, indexing strategies, and distributed query execution plans** for high-volume datasets, improving query performance by 38% in production workloads
- Built production-ready Python LLM application with tool integrations, retrieval components, and multi-turn context summarization — improving response accuracy by 43% through iterative prompt pipeline design, component testing, and stakeholder feedback cycles
- Developed **asynchronous backend workflows** for real-time, fault-tolerant processing of longitudinal health data with focus on scalability

Research Assistant - CHI AI in Infodemic Management

January 2025 - May 2025

- Designed and deployed scalable LLM and NLP pipelines in Python on GCP, applying modular component architecture and prompt engineering techniques to reduce processing latency by 41% across high-volume enterprise datasets
- Built distributed data ingestion pipelines using **Kafka** and **Neo4j** Streams for real-time, high-throughput processing
- Designed and developed **production-grade REST APIs using FastAPI** to serve AI models, implementing validation, monitoring, and automated deployment workflows to ensure **scalable integration** across distributed systems
- Enabled large-scale data processing across social media streams using **distributed systems and large-scale processing** principles

Swift Mobil Software Solutions Provider

July 2023 - October 2023

Data Analyst

Mumbai, IN

- Built interactive **Power BI** dashboards for **logistics** and **mobility KPIs**, reducing reporting time by **40%** and improving operational visibility
- Processed and optimized large-scale datasets using **Python** and **PySpark** to identify failure and performance issues in distributed systems
- Developed **SQL** and **graph queries** to map dependencies, analyze operations, detect data issues, speeding root-cause analysis by **26%**

Pscope Technologies Pvt. Ltd.

January 2023 - June 2023

Data Analyst

Mumbai, IN

- Developed and maintained **20+ Power BI** dashboards across Sales, Finance, and Operations for enterprise clients, leveraging **SQL** and **DAX**, enabling KPI tracking and data-driven decisions
- Automated recurring **Excel**-based workflows using **VBA**, implementing validation and scheduling logic, reducing report preparation time by **42%** and improving consistency

Stu/Dio at Illinois

August 2025 - Present

Project Manager

Champaign, USA

- Led cross-functional team of **6** engineers and designers to deliver **AI-driven product features** for DeepCover game, leveraging feedback and performance insights for feature prioritization
- Tracked **sprint** progress, delivery metrics, and **QA issues** through **JIRA**-based workflows, managing backlog and issue resolution

PROJECTS

Retail Demand Forecasting

March 2025

[View Project](#)

- Analyzed FMCG retail sales data across product categories and store locations to identify demand trends, seasonal patterns, and inventory drivers
- Built and evaluated forecasting models, ARIMA, SARIMA, and regression, using RMSE and MAPE, deriving actionable insights on pricing, promotions, and stock behavior for inventory planning
- Translated analytical findings into business recommendations, reducing simulated stockout and overstock risk across seasonal demand cycles